Oct 25 — 2 papers

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Vehicle-State-Aware Confidence Estimation for Direct Wind Sensing on a Multirotor

SKIM

Sri Arjun Senthil Kumar, April Zi Qian Chan, Chen-Hua Yeow

A drone-mounted anemometer's readings get down-weighted whenever the drone is moving fast horizontally or vertically, and that is essentially the whole idea.

PROBLEM A wind sensor mounted on a multirotor is corrupted by the vehicle's own motion and rotor wash. Existing estimators usually take every reading at face value, or throw readings out with hand-set rules, instead of grading how trustworthy each one is.

METHOD They build a confidence score from just two inputs, horizontal speed and vertical speed of the vehicle. That score scales the gain of a two-dimensional wind vector estimator, so readings taken while the drone is moving barely update the wind estimate and readings taken in hover update it normally.

RESULTS The two-feature score separates reliable from unreliable readings with an ROC-AUC of 0.995 and flags 98.84 percent of translation readings as unreliable. Median estimator gain drops from 0.207 to 0.064 during translation and stays at 0.151 in hover, tested on translation, vertical motion and one held-out autonomous waypoint flight. The abstract compares against no external baseline.

CAVEAT The near-perfect AUC looks circular: if reliability labels come from flight phase, then vehicle speed predicts them almost by definition. The abstract also reports lower gain, not lower wind estimation error against an independent reference such as a ground mast, so we don't know whether the final wind estimate actually gets better. Everything comes from one airframe and one sensor.

WHY IT MATTERS This is outside your area. It involves no inter-vehicle communication, no negotiation, no platooning and no multi-agent planning. The only loose link is the general idea of using the sender's own motion state to weight how much to trust its data, which you could apply to incoming intent or trajectory messages, but the paper offers nothing you can transfer directly.

arXiv:2610.25688v1 PDF incremental

DexGraspFlow: Topology-Aware SE(3) Flow Matching for Cross-Embodiment Dexterous Grasp Generation

SKIM

Bin Zhou et al.

A flow-matching grasp generator that predicts a pose for every link of a robot hand, uses the hand's kinematic tree to bias attention, and transfers across three different dexterous hands.

PROBLEM Grasp generators trained on one hand do not transfer well to hands with different geometry, joint layouts or finger counts. The authors argue that earlier cross-embodiment methods model only spatial relations between links and ignore how the links are connected.

METHOD They run conditional flow matching over the rigid pose of each link, learning translation and rotation velocity fields jointly. A dual-stream diffusion transformer handles link-to-link and link-to-object attention. Attention is biased by distances in the kinematic tree. Joint angles are recovered afterwards by inverse kinematics, and a guidance term applied at inference time steers palm orientation without retraining.

RESULTS Across three robotic hands and 30 object-hand pairs, they report a 95.33 percent average simulation success rate and roughly 49 percent lower generation latency, inverse kinematics included, than an unnamed state-of-the-art baseline. They also show zero-shot transfer to seven finger-removal variants and 91 percent success over 100 real-robot trials.

CAVEAT The real-robot model was trained on demonstrations from the deployed hand itself, so the 91 percent figure does not test cross-embodiment transfer. The finger-removal variants are subsets of hands already seen in training, not new kinematic designs. The abstract also never names the baseline and gives no ablation of the kinematic-tree attention bias.

WHY IT MATTERS This is outside your area. It is about single-robot dexterous manipulation, with no connected vehicles, no communication, no negotiation and no multi-agent planning. The only loose link is the idea of encoding graph structure as an attention bias, which you can get from better-matched graph-transformer work on vehicle interaction.

arXiv:2610.26962v1 PDF incremental